

**University of North Carolina at Charlotte
Department of Electrical and Computer
Engineering**

**Homework Report #4
ECGR 4106 - Real-Time Machine Learning**

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GitHub: [pGovM/introToDL](#)

1 Hyper Parameters

These hyperparameters were shared in all 3 problems

- Hidden Size: 128
- Epochs: 50
- Learning Rate: 1e-4
- Maximum Length of Sentence: 100
- Batch Size: 1 (problems 3 & 4), 16 (problems 1 & 2)
- Train/Val Split: 80/20
- Random Seed: 42

Problem 1

Objective: Train and validate a transformer model for learning the given sequence, with sequence lengths of 10, 20, and 30. Report and compare the model metrics against the RNN-based approach in the previous homework.

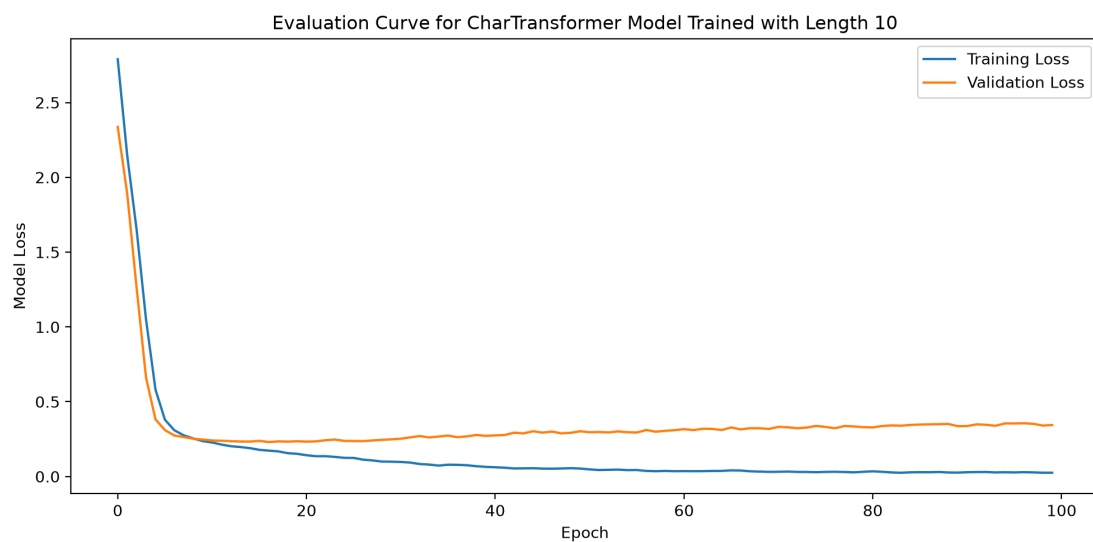


Figure 1: **Evaluation Curves for Transformer Model Trained with Sequence Length of 10**

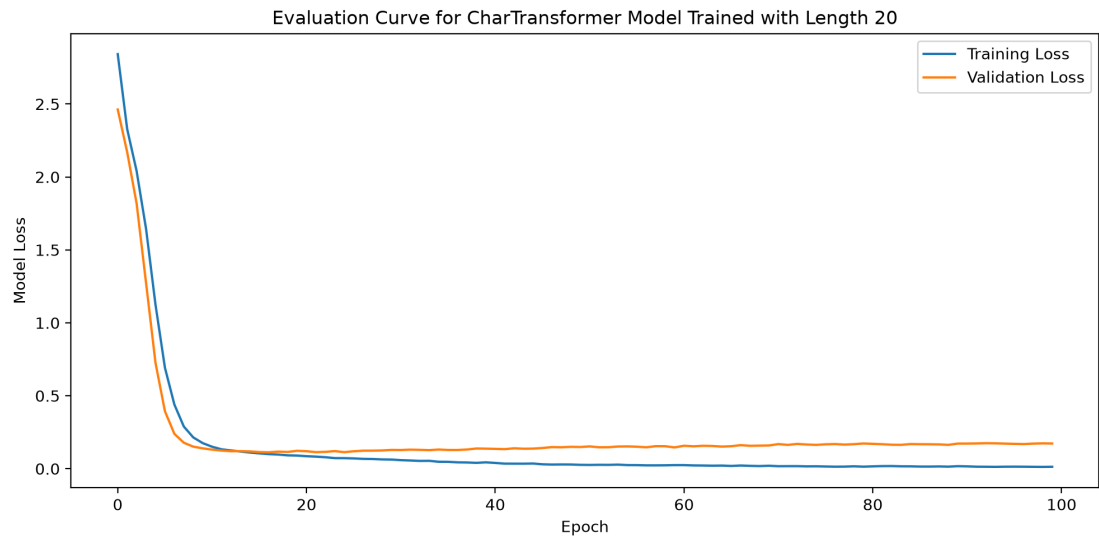


Figure 2: Evaluation Curves for Transformer Model Trained with Sequence Length of 20

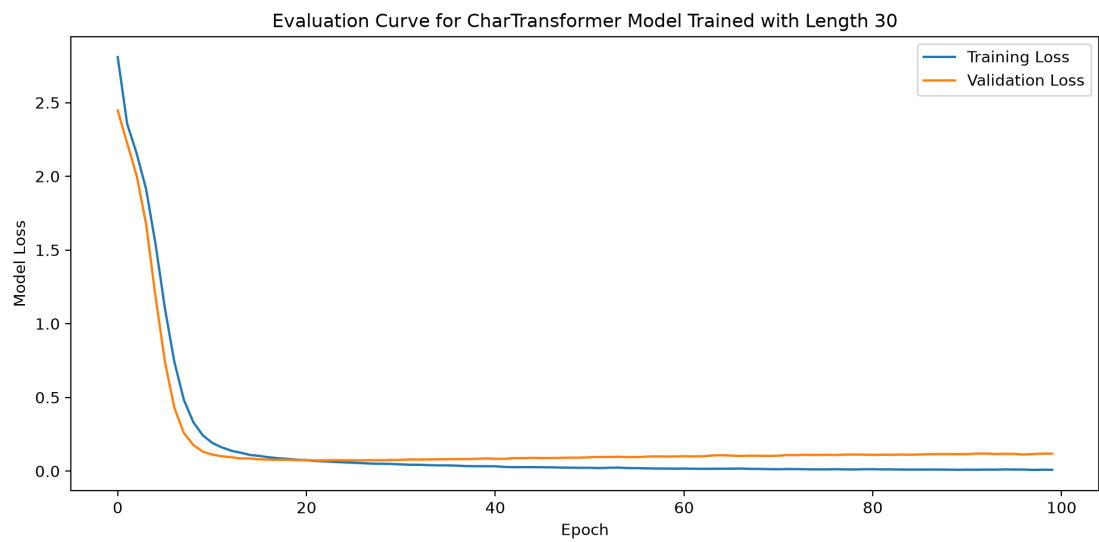


Figure 3: Evaluation Curves for Transformer Model Trained with Sequence Length of 30

Training Loss

Table 1: Final Training Loss

Sequence Length	Model	Transformer	RNN	LSTM	GRU
10		0.1215	0.0906	0.0888	0.0630
20		0.0965	0.1039	0.1566	0.0992
30		0.0634	0.1126	0.1592	0.0419

Validation Loss

Table 2: **Final Validation Loss**

Sequence Length	Model	Transformer	RNN	LSTM	GRU
10		0.2249	2.540	2.557	2.571
20		0.1087	2.694	2.297	2.495
30		0.0722	2.740	2.472	2.697

Validation Accuracy

Table 3: **Validation Accuracy (%)**

Sequence Length	Model	Transformer	RNN	LSTM	GRU
10		94.35	49.26	44.86	50.73
20		97.00	50.31	49.47	52.42
30		97.99	47.99	47.57	49.26

Model Complexity

Table 4: **Model Complexity (FLOPs)**

Sequence Length	Model	Transformer	RNN	LSTM	GRU
10		10.56M	337.2k	1.337M	1.005M
20		21.13M	668.8k	2.668M	2.005M
30		31.69M	1M	3.999M	3.005M

Number of Parameters

Table 5: **No. of Params**

Sequence Length	Model	Transformer	RNN	LSTM	GRU
10		1.06M	44.59k	143.6k	110.6k
20		1.06M	44.59k	143.6k	110.6k
30		1.06M	44.59k	143.6k	110.6k

Model Size

Table 6: **Model Size (MB)**

Sequence Length	Model	Transformer	RNN	LSTM	GRU
10		7.0099	0.1701	0.5480	0.4220
20		7.0099	0.1701	0.5480	0.4220
30		7.0099	0.1701	0.5480	0.4220

Training Time

Table 7: **Training Time (s)**

Sequence Length	Model	Transformer	RNN	LSTM	GRU
10		7.0099	0.56	1.10	0.79
20		7.0099	0.41	1.76	1.22
30		7.0099	0.63	2.37	1.70

Problem 2

Objective: Build a transformer model for the tiny Shakespeare dataset with sequence lengths of 20, 30, and 50 while exploring transformer architectures with 1, 2, and 4 transformer blocks, combined with 2 and 4 heads. Report the resulting metrics.

Problem 3

Objective: Develop a transformer-based encoder-decoder architecture for English-to-French translation, explore the same transformer architectures of 1, 2, and 4 transformer blocks with 2 and 4 heads. Report standard model metrics combined with traditional sequence accuracy and validation bleu-4 score based on the validation set. Compare your best-performing transformer configuration against the Homework 3 RNN-based models, across all the previously mentioned metrics.

Problem 4

Objective: Repeat the same exercise as problem 3 but translate from French to English using the same dataset.